Project Design Phase-II

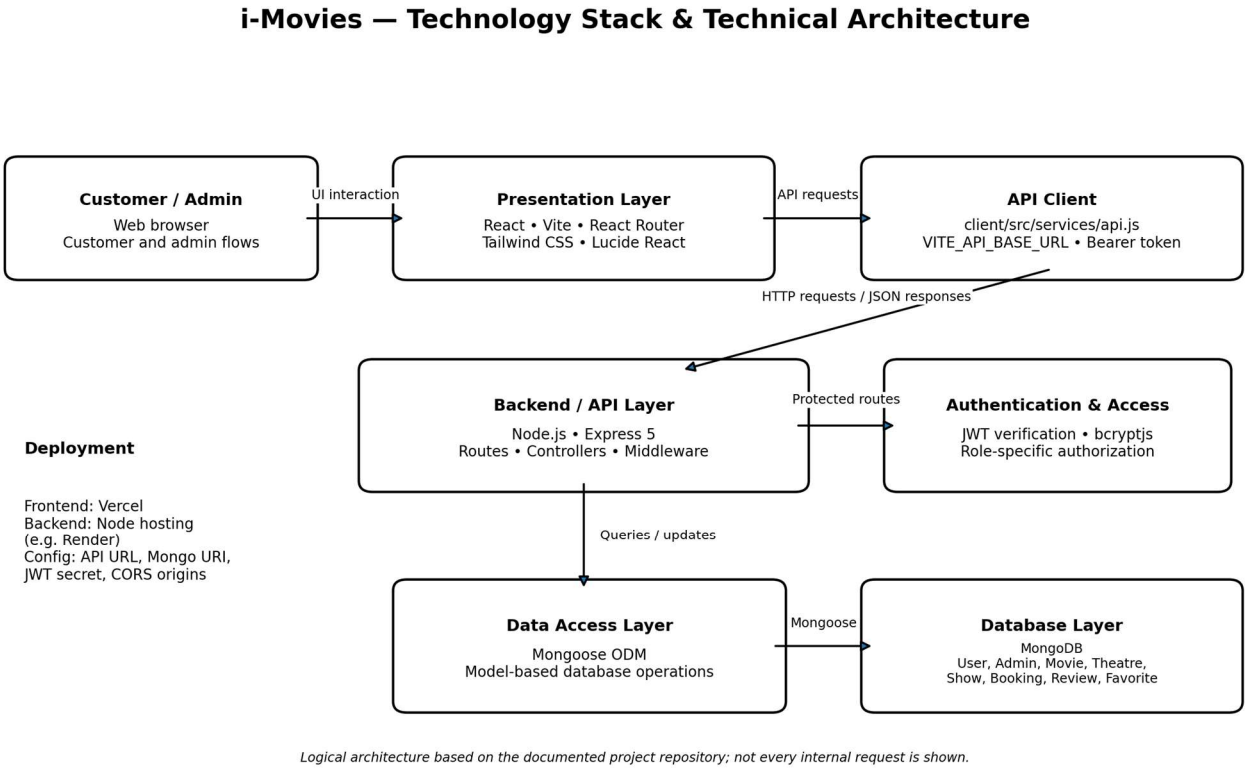
Technology Stack (Architecture & Stack)

Date	22-10-2026
Team ID	
Project Name	i-Movies – Movie Ticket Booking System
Maximum Marks	4 Marks

Technical Architecture

i-Movies follows a layered full-stack architecture. The React frontend communicates with a Node.js/Express API, which applies authentication and authorization checks and accesses MongoDB through Mongoose models. The frontend and backend are deployed separately, so the API base URL and CORS configuration must match the deployed origins.

Technology Architecture Diagram



Logical architecture based on the documented project repository; not every internal request is shown.

Table 1: Components & Technologies

S. No.	Component	Description	Technology
1	User Interface	Customer-facing and administrator web interface; routing and responsive styling.	React, Vite, React Router, Tailwind CSS, Lucide React
2	API Client	Centralized frontend API requests; reads VITE_API_BASE_URL and attaches bearer token when available.	JavaScript module: client/src/services/api.js
3	Application Logic	REST API endpoints, route handling, controllers, middleware, validation, and error responses.	Node.js, Express 5
4	Authentication & Authorization	Password hashing, token-based authentication, token verification, and role checks for protected routes.	JWT, bcryptjs, Express middleware
5	Database & Data Access	Persistent application data and schema/model operations.	MongoDB, Mongoose
6	Core Data Models	Accounts and movie-ticket-booking domain data.	User, Admin, Movie, Theatre, Show, Booking, Review, Favorite
7	Deployment	Frontend and backend are hosted separately; environment variables connect them securely.	Vercel frontend; Node.js backend hosting such as Render

Table 2: Application Characteristics

S. No.	Characteristic	Description	Technology / Implementation
1	Open-source frameworks	Frontend and backend use established JavaScript ecosystem tools.	React, Vite, Node.js, Express, Tailwind CSS
2	Layered architecture	Presentation, API/business logic, and data-access layers have separate responsibilities.	React frontend; Express API; Mongoose models
3	Authentication security	Passwords are hashed and protected endpoints verify JWTs and user roles.	bcryptjs, JWT, authentication middleware
4	Booking integrity	Seat reservation uses conditional updates to reduce double-booking; failed booking creation can trigger rollback.	Express controllers and MongoDB/Mongoose operations
5	Configuration & deployment	Environment configuration controls API URL, database connection, JWT secret, and allowed CORS origins.	VITE_API_BASE_URL, MONGO_URI, JWT_SECRET, CORS_ORIGINS

References

i-Movies project repository README: https://github.com/KunalGore2005/IMovies-movie_ticket_booking_system